

Alessandro Zito

Italy: +39 349-461-3584 | US: +1 984-528-0287 | azito@hsph.harvard.edu | [Linkedin](#) | [GitHub](#) | [Website](#)

HIGHLIGHT

Experienced statistician working as a postdoctoral fellow at Harvard University, Department of Biostatistics, and the Dana-Farber Cancer Institute. Current work includes dimensionality reduction methods for genomic and clinical data. Ph.D. graduate in Statistics from Duke University. Great experience in applied and methodological research, with particular interests in DNA-related data, nonparametric methods, and high-dimensional statistics.

Extensive research collaborations with scientists and physicians outside the statistical field, and with data scientists in industry (Generali and Google).

EXPERIENCE

Postdoctoral researcher, Biostatistics

Boston, MA, USA

Harvard School of Public Health and Dana-Farber Cancer Institute

Sep 2023 – current

- Working with Prof. [Jeff Miller](#) Prof. [Giovanni Parmigiani](#) and Dr. [Mehmet K. Samur](#) on methods to understand patterns of mutations in the DNA and RNA of cancer cells, and their relationship with clinical outcomes.
- Areas of research: Machine Learning, High-dimensional statistics, Dimensionality reduction methods, regression methods, cancer genomics, mutational signature analysis

Intern Data Scientist

San Bruno, CA, USA

Google LLC, Youtube Data Science team

May 2022 – Aug 2022

- Implemented Bayesian hypothesis testing methods to evaluate platform experiments
- Designed a methodology to construct proxy metrics to predict the long-term outcome of experiments, using multivariate Bayesian optimization algorithms and Gaussian processes.
- Extended the collaboration after the internship to publish the methods. See paper [here](#).

Intern Data Scientist

Milan, Italy

Generali Italia S.p.A., Advanced Analytics Solutions team.

Jun 2018 – Jul 2019

- Developed and implemented a Bayesian bandit algorithm recommendation system in Python via a Google Cloud Platform API to optimize a life insurance policy marketing campaign
- Implemented a drivers profiling model, migrated from the partner society MyDrive Solutions, using Python

EDUCATION

Duke University

Durham, NC, USA

Ph.D. in Statistical Science.

Aug 2019 – Sep 2023

- Supervisor: Professor [David B. Dunson](#). Co-supervisor: Professor [Tommaso Rigon](#)
- Research interests: Bayesian Methods, Bayesian Nonparametrics, Species Sampling Models, Ecology
- Funded by project [Lifeplan](#), a European Research Council grant aiming at mapping global trends in biodiversity.
- Dissertation: *Ecological modeling via Bayesian nonparametric species sampling priors* [Link](#) [Slides](#) .
- Recipient of the [2024 Savage Award](#) for Outstanding Dissertation in Applied Methodology

Bocconi University

Milan, Italy

Master's of Science in Economics and Social Sciences

Sep 2016 – Mar 2019

- Graduation grade: 110 cum laude / 110
- Thesis: *Conjugate Fractional Factorial Thompson Sampling via unified skew-normal*, with Prof. [Daniele Durante](#)

Bocconi University

Milan, Italy

Bachelor's of Science in Economics and Social Sciences

Sep 2013 – Oct 2016

- Graduation grade: 110 cum laude / 110
- Exchange program at The University of Chicago, Department of Economics, Fall 2015

AVAILABLE PREPRINTS

Xue, C., **Zito, A.** and Miller, J. W. (2024), Improved control of Dirichlet location and scale near the boundary , [ArXiv:2410.13050](#), *invited reject and resubmit*.

Zito, A. and Miller, J. W. (2024), Compressive Bayesian non-negative matrix factorization for mutational signature analysis, [ArXiv:2404.10974](#), *under review*.

PUBLICATIONS

- Lee, C. J., **Zito, A.**, Sang, H. and Dunson, D. B. (2025), Logistic-beta processes for dependent random probabilities with beta marginals, *Bayesian Analysis*. Advanced Publications 1-25. DOI:10.1214/25-BA1541
- Zito, A.**, Greaves, D., Soriano, J. and Richardson, L. (2025), Pareto Optimal Proxy Metrics, *Applied Stochastic models in Business and Industry*, 41: e70003. DOI:10.1002/asmb.70003
- Zito, A.**, Rigon, T. and Dunson, D. B. (2024), Bayesian nonparametric modelling of latent partitions via Stirling-gamma priors, *Bayesian Analysis*. Advanced Publications 1-24. DOI:10.1214/24-BA1463.
- Zito, A.**, Rigon, T., Ovaskainen, O. and Dunson, D. B. (2023), Bayesian modeling of sequential discoveries, *Journal of the American Statistical Association (T&M)*, 118:544, 2521-32, DOI:10.1080/01621459.2022.2060835
- Zito, A.**, Rigon, T. and Dunson, D. B. (2023), Inferring taxonomic placement from DNA barcoding aiding in discovery of new taxa. *Methods in Ecology and Evolution*. 14, 529–42. DOI:10.1111/2041-210X.14009

CONFERENCE PROCEEDINGS AND DISCUSSIONS

- Zito, A.**, Xue, C., and Miller, J. W. (2025). Accounting for Transcriptional Asymmetries in Mutational Signature Analysis Using Compressive Bayesian Non-negative Matrix Factorization. In: di Bella, E., Gioia, V., Lagazio, C., Zaccarin, S. (eds) *Statistics for Innovation IV. SIS 2025. Italian Statistical Society Series on Advances in Statistics*. Springer, Cham. [Link](#)
- Zito, A.** and Miller, J. W. (2024). Discussion to “Sparse Bayesian factor analysis when the number of factors is unknown” by Fruhwirth-Schnatter, S., Hosszejni, D., and Lopes, F. L., *Bayesian Analysis*. [\[publication link\]](#)
- Zito, A.**, Rigon, T. and Dunson, D. B. (2021). Modelling of accumulation curves through Weibull survival functions. In *Book of Short Papers of the Italian Statistical Society 2021* (Perna, C., Salvati, N. and Schirripa Spagnolo, F., editors). ISBN: 9788891927361.

MANUSCRIPTS IN PREPARATION

- Zito, A.**, Parmigiani, G., and Miller, J. W. (2025). Topography-dependent mutational signature analysis via Poisson process factorization. *expected submission: November 2025*
- Orsholm, J.*, **Zito, A.***, Somervuo, P., Braga M, Chazlot, N, Roslin, T. and Furneaux, B. (2025). Discovering the unseen: a performance comparison of taxonomic classification methods under unknown DNA barcodes. *expected submission: October 2025*
- Zito, A.**, Rigon, T., Niittynen, P., Hebert, P. D. N., Zakharov, E. V., Ratnasingham, R., the iBOL consortium, Ovaskainen, O. and Dunson, D. B. (2025). Covariate-dependent biodiversity measures via the Hubbell regression. *expected submission: December 2025*
- Zito, A.**, Parmigiani, G., Miller, J. W. and Samur, M. (2025+). Joint analysis of mutational process in DNA and RNA from multiple myeloma.

SOFTWARE

- Zito, A.** (2023) **CompressiveNMF**: Bayesian compressive non-negative matrix factorization software [GitHub](#)
- Zito, A.** (2023) **ConjugateDP**: Random number generator for the Stirling-gamma distribution. [GitHub](#)
- Zito, A.** (2022) **BayesANT**: Bayesian nonparametric taxonomic classifier for DNA barcoding sequences. [GitHub](#)
- Zito, A.**, Rigon, T. (2021) **BNPvegan**: Bayesian nonparametric methods for ecology. [GitHub](#)

SKILLS

IT: R, Rcpp (Fluent), Python, Keras, Tensorflow, Git, Cluster computing via SLURM, SQL (Proficient)
Languages: Italian (Native), English (Fluent), Spanish (Proficient)

PROFESSIONAL ACTIVITIES

Board member of [j-ISBA](#), the junior section of the International Society for Bayesian Analysis. Elected as Treasurer. Current duties involve organizing the 2025 and 2025 Blackwell-Rosembluth awards, organizing BAYSM2026 in Chiba, Japan, and expanding the outreach of j-ISBA to lower income countries.

Referee activity: *Electronic Journal of Statistics, Bayesian Analysis, Computational Statistics & Data Analysis, Ecography, Philosophical transactions of the Royal Society A.*

AWARDS

BNP14 Travel Award
ISBA 2024 Savage Award for Outstanding Dissertation in Bayesian Applied Methodology
ISBA 2022 Travel Award
BEST Award for Ph.D. student research for the 2021/2022 academic year
Ph.D. Teaching Assistant of the Year Award for the 2020/2021 academic year
Best Student/Postdoc Contributed Paper Award at ISBA 2021 World meeting
Bocconi Merit Award scholarship for the 2016/2017 and 2017/2018 academic years

ORAL PRESENTATIONS AND TALKS

JSM2025, Nashville, TN – *Ecological Modeling via Bayesian Nonparametric Species Sampling Priors* (Invited talk, The Savage Dissertation Award)
BNP14, Los Angeles, CA – *Topography-dependent mutational signature analysis via Poisson process factorization* (Contributed talk)
ENAR2025, New Orleans, LA – *Compressive Bayesian non-negative matrix factorization for mutational signature analysis* (Contributed talk)
Brown University, Data Science Initiative, Providence, RI, USA – *Compressive Bayesian non-negative matrix factorization for mutational signature analysis* (Invited seminar)
Cattolica University of the Sacred Heart, Milan, Italy – *Compressive Bayesian non-negative matrix factorization for mutational signature analysis* (Invited seminar)
ICSDS2024, Nice, France – *Compressive Bayesian non-negative matrix factorization for mutational signature analysis* (Contributed talk)
ISBA2024, Venice, Italy – *Compressive Bayesian non-negative matrix factorization for mutational signature analysis* (Invited talk)
ISBA2024, Venice, Italy – *Bayesian nonparametric modeling of latent partitions via Stirling-gamma priors* (Poster)
CMstat2023, Berlin, Germany – *Bayesian nonparametric modeling of latent partitions via Stirling-gamma priors* (Invited talk)
EcoSta2023, Tokyo, Japan – *Bayesian nonparametric modeling of latent partitions via Stirling-gamma priors* (Invited talk)
NESS2023, Boston, MA – *Bayesian nonparametric modeling of latent partitions via Stirling-gamma priors* (Invited talk)
CMstat2022, London, UK – *Inferring taxonomic placement from DNA barcoding aiding in discovery of new taxa* (Invited talk)
BNP13, Puerto Varas, Chile – *Bayesian nonparametric modeling of latent partitions via Stirling-gamma priors* (Poster)
ISBA2022, Montreal, Canada – *Inferring taxonomic placement from DNA barcoding aiding in discovery of new taxa* (Poster)
Luomos SYKE Machine Learning Seminar II (Virtual) – *Taxonomic classification methods for DNA sequences* (Contributed talk)
ISBA2021 (Virtual) – *Bayesian modeling of sequential discoveries* (Contributed talk)
ISBA2021 (Virtual) – *Inferring taxonomic placement from DNA barcoding aiding in discovery of new taxa* (Contributed talk)

SESSIONS ORGANIZED AND CHAIRED

BAYSM2025. Member of the scientific and organizing committee.
JSM2025, Nashville, TN – *Advances in Bayesian Factorization Methods in Genomics and Medicine* (Invited session). Speakers: Adolphus Wagala (Dana-Farber Cancer Institute), Jonathan Huggins (Boston University), Sarah Urbut (Broad Institute), Niccolo Anceschi (Duke University). Discussion by Peter Carbonetto (University of Chicago).
ISBA2024, Venice, Italy – *Bayesian Methods in Ecology* (Invited session). Speakers: Bokgyeong Kang (Duke University), Michele Peruzzi (University of Michigan), Narmadha M. Mohankumar (Pacific Northwest National Lab).

OTHER ACHIEVEMENTS

Painter. Personal exhibitions at Triennale di Milano museum of arts and “Spazio Arte Tirabassi” art gallery.
Winner of the 2018 “Looking for Art” contest in Milan.
Former semi-professional actor at “Proxima Res”, Milan